

Iron-Catalyzed Vinylzincation of Terminal Alkynes

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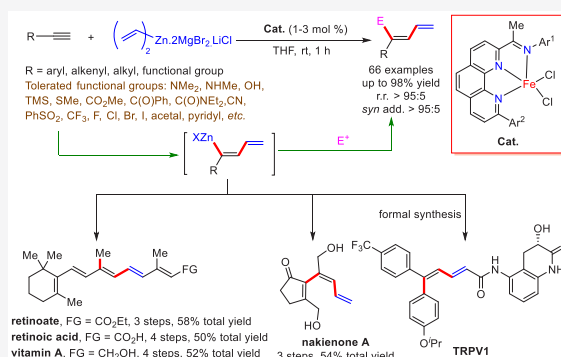


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ABSTRACT: Organozinc reagents are among the most commonly used organometallic reagents in modern synthetic chemistry, and multifunctionalized organozinc reagents can be synthesized from structurally simple, readily available ones by means of alkyne carbozincation. However, this method suffers from poor tolerance for terminal alkynes, and transformation of the newly introduced organic groups is difficult, which limits its applications. Herein, we report a method for vinylzincation of terminal alkynes catalyzed by newly developed iron catalysts bearing 1,10-phenanthroline-imine ligands. This method provides efficient access to novel organozinc reagents with a diverse array of structures and functional groups from readily available vinylzinc reagents and terminal alkynes. The method features excellent functional group tolerance (tolerated functional groups include amino, amide, cyano, ester, hydroxyl, sulfonyl, acetal, phosphono, pyridyl), a good substrate scope (suitable terminal alkynes include aryl, alkenyl, and alkyl acetylenes bearing various functional groups), and high chemoselectivity, regioselectivity, and stereoselectivity. The method could significantly improve the synthetic efficiency of various important bioactive molecules, including vitamin A. Mechanistic studies indicate that the new iron-1,10-phenanthroline-imine catalysts developed in this study have an extremely crowded reaction pocket, which promotes efficient transfer of the vinyl group to the alkynes, disfavors substitution reactions between the zinc reagent and the terminal C–H bond of the alkynes, and prevents the further reactions of the products. Our findings show that iron catalysts can be superior to other metal catalysts in terms of activity, chemoselectivity, regioselectivity, and stereoselectivity when suitable ligands are used.



INTRODUCTION

Organometallic reagents are powerful tools for modern synthetic chemistry. Especially, organozinc reagents readily undergo diverse transformations (e.g., Negishi cross-coupling reactions and addition reactions with unsaturated compounds) and are compatible with a variety of functional groups. Moreover, the abundance, low cost, and good biocompatibility of zinc have resulted in organozinc reagents becoming some of the most commonly used reagents for the synthesis of pharmaceuticals, agrochemicals, cosmetics, and functional materials.¹ Organozinc reagents have traditionally been prepared by reactions of zinc metal with organohalides or by metal-exchange reactions between zinc salts and highly active magnesium (Grignard) reagents or lithium reagents.¹ Unfortunately, the structural and functional group diversity of the reagents accessible by these methods is seriously limited. Therefore, the development of more-versatile synthetic methods is a challenge that must be overcome for expansion of the applications of organozinc reagents.

Carbozincation reactions of alkynes, which simultaneously introduce an organic moiety and a zinc atom at opposite ends of a carbon–carbon triple bond, is a straightforward method for preparing multifunctionalized organozinc reagents from structurally simple ones. Catalytic alkyne carbozincation can be accomplished under mild conditions with controllable chemo-

selectivity, regioselectivity, and stereoselectivity. Although this promising method is attracting increasing attention,² two problems hamper its further development. First, although methods for carbozincation of internal alkynes are well established,³ reactions of terminal alkynes remain challenging because the highly acidic proton of these compounds can react with basic zinc reagents and destroy active catalysts. To our knowledge, only aryl acetylenes have successfully been used in carbozincation reactions.^{3e,4} Second, the organic groups (e.g., alkyl and aryl) typically introduced by alkyne carbozincation do not readily undergo subsequent functionalization, which limits the utility of this reaction.

We hypothesized that if a zinc atom and a readily transformable organic moiety could be simultaneously introduced at the ends of a terminal alkyne by means of carbozincation, the resulting doubly functionalized alkenylzinc would be of great synthetic utility (Scheme 1A). Specifically, we reasoned that catalytic vinylzincation of terminal alkynes

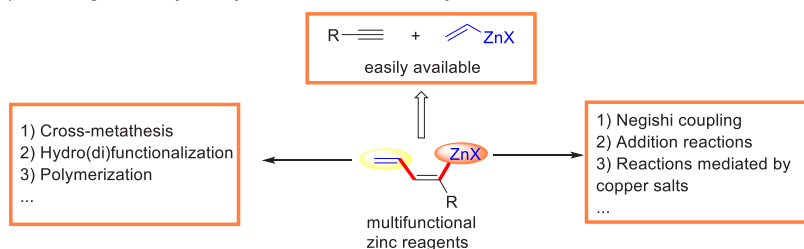
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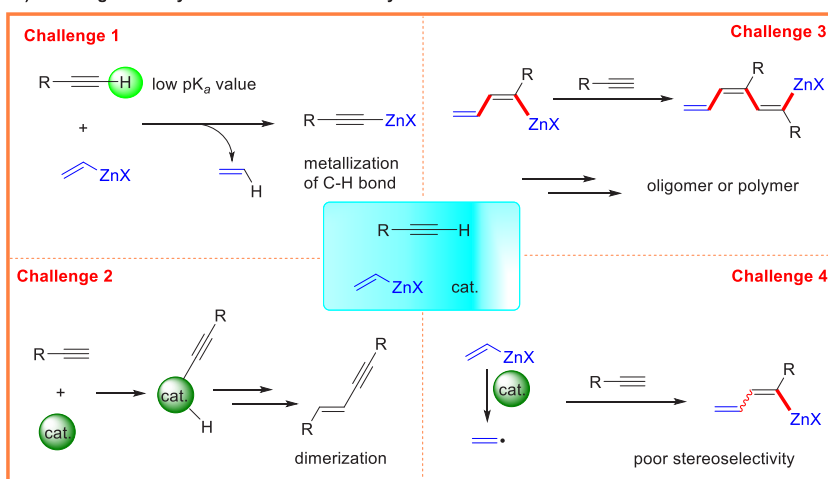


Scheme 1. Catalytic Vinylzincation of Terminal Alkynes

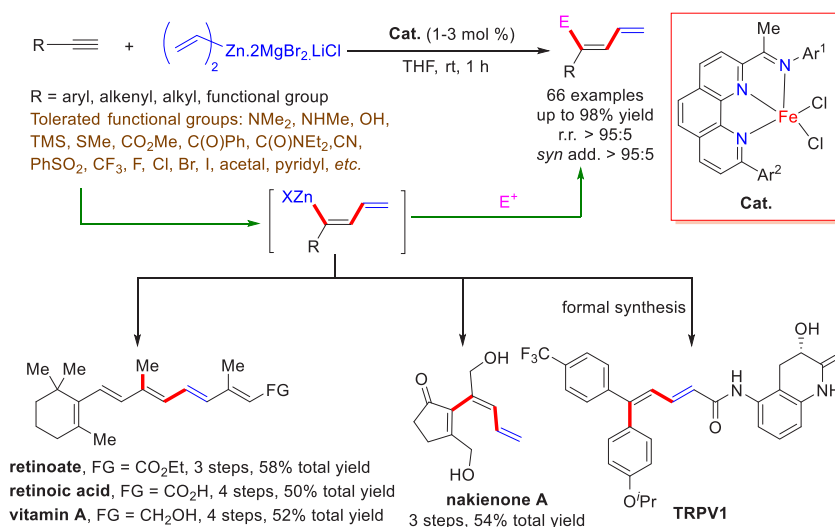
A) The design of catalytic vinylzincation of terminal alkynes



B) Challenges in vinylzincation of terminal alkynes



C) This Work: Fe-catalyzed vinylzincation of terminal alkynes



might be useful for this purpose because vinyl groups readily undergo a variety of reactions, such as olefin metathesis, hydrogen functionalization, and olefin polymerization.^{5–8} However, to achieve the desired transformation, we would have to overcome the following challenges (Scheme 1B). First, as stated above, the highly acidic proton of terminal alkynes can quench the reaction or lower its chemoselectivity.^{2a,4} Second, terminal alkynes readily undergo dimerization through oxidative addition in the presence of transition-metal catalysts.⁹ Third, the alkenylzinc compounds generated by vinylzincation of alkynes can react with the starting alkyne to form alkyne oligomers or polymers.¹⁰ Fourth, zinc reagents readily generate radicals under the catalysis of transition metals,^{11,12} which makes controlling the stereoselectivity of the reaction difficult.

As part of our ongoing research on iron-catalyzed reactions,¹³ we herein report the development of a method for vinylzincation of terminal alkynes catalyzed by well-defined iron complexes bearing newly developed tridentate 1,10-phenanthroline-imine ligands (Scheme 1C). The method exhibited impressive functional group tolerance and substrate scope: aryl, alkenyl, and alkyl acetylenes and various functionalized terminal alkynes smoothly afforded the desired alkenylzinc products in high yields with excellent chemo-, regio-, and stereoselectivity. Substrates with amino, amido, cyano, ester, hydroxy, sulfonyl, acetal, phosphono, and pyridyl functional groups were suitable, and the method was successfully used to modify various pharmaceuticals and for concise syntheses of some bioactive molecules, including

Table 1. Iron-Catalyzed Vinylzincation of Terminal Alkyne 1a: Optimizing Reaction Conditions^a

catalyst	conv (%)	2aa (yield, %)	3aa (yield, %)	4aa (yield, %)
1 none	45	no	no	no
2 L1	40	no	no	no
3 FeCl ₂	70	8 (NA)	5 (NA)	6 (NA)
4 C1	> 95	99 (> 95)	trace	trace
5 C2	> 95	90 (> 95)	trace	trace
6 ^b C2	> 95	91 (> 95)	trace	trace
7 ^c C2	> 95	92 (> 95)	trace	trace
8 C3	> 95	99 (> 95)	trace	trace
9 C4	> 95	83 (> 95)	12 (> 95)	trace
10 C5	> 95	73 (> 95)	trace	trace
11 C6	> 95	96 (> 95)	trace	trace
12 C7	> 95	87 (> 95)	8 (NA)	trace
13 C8	> 95	41 (> 95)	54 (> 95)	trace
14 C9	60	trace	trace	trace
15 C10	> 95	54 (> 95)	9 (NA)	trace
16 C11	> 95	68 (> 95)	12 (> 95)	trace
17 C12	92	23 (> 95)	trace	27 (> 95)

C1 Ar = 2,6-Me₂C₆H₃

C3

C4 R = H

C7 Ar = 2,6-Me₂C₆H₃

C9 Ar = 3,5-t-Bu₂C₆H₃

C11 Ar = 2,6-Pr₂C₆H₃

C12

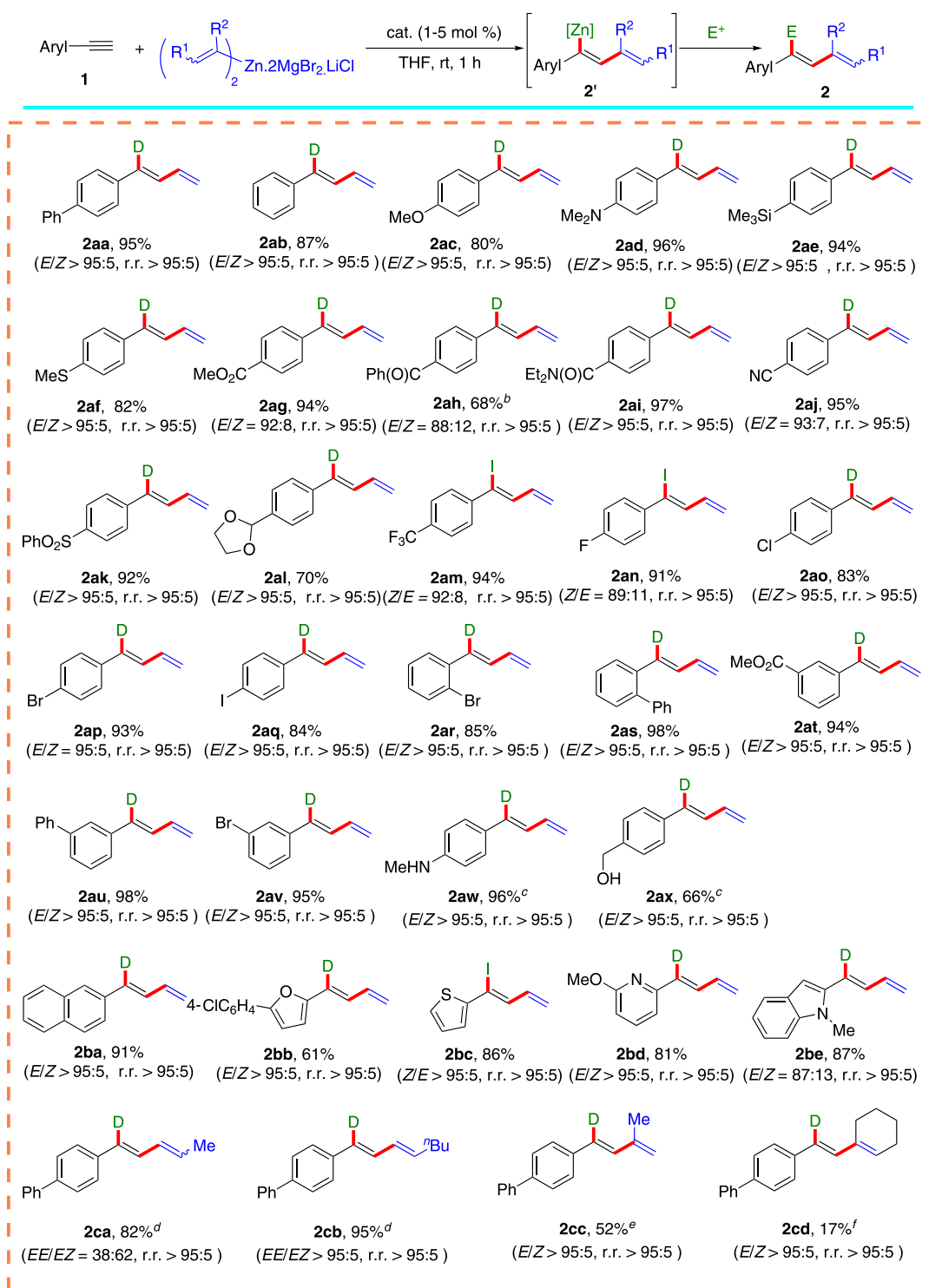
^aReaction conditions: **1** (0.3 mmol), divinylzinc (0.33 mmol), catalyst (1 mol %), THF (3.0 mL), rt, 1 h. Conversions and yields were determined by ¹H NMR using 1,3,5-trimethoxybenzene as an internal standard. The data in parentheses is the amount of ²H-incorporation. NA = Not analyzed. The catalysts used in the text are all preprepared. ^bUsed 2 mol % of (*p*-Tol)MgBr as additive. ^cUsed 1 mol % of di(1-phenylvinyl)zinc as additive.

vitamin A. The rigid phenanthroline backbone and bulky aryl substituents of the ligands played crucial roles in ensuring the yield and selectivity of the vinylzincation reactions.

RESULTS AND DISCUSSION

We began by using 4-phenylphenylacetylene (**1aa**) as a model substrate in reactions with a divinylzinc reagent at room temperature in THF (Table 1). The desired reaction failed to proceed in the absence of a catalyst/ligand combination (entry 1) or in the presence of only ligand **L1** (the ligand shown in catalyst **C1**, entry 2), but alkynylzinc, formed through proton-zinc exchange between alkyne **1aa** and divinylzinc, was detected as major byproduct. When FeCl₂ was used as the catalyst, substrate conversion was incomplete, and the reaction showed poor selectivity, producing a variety of carbocation

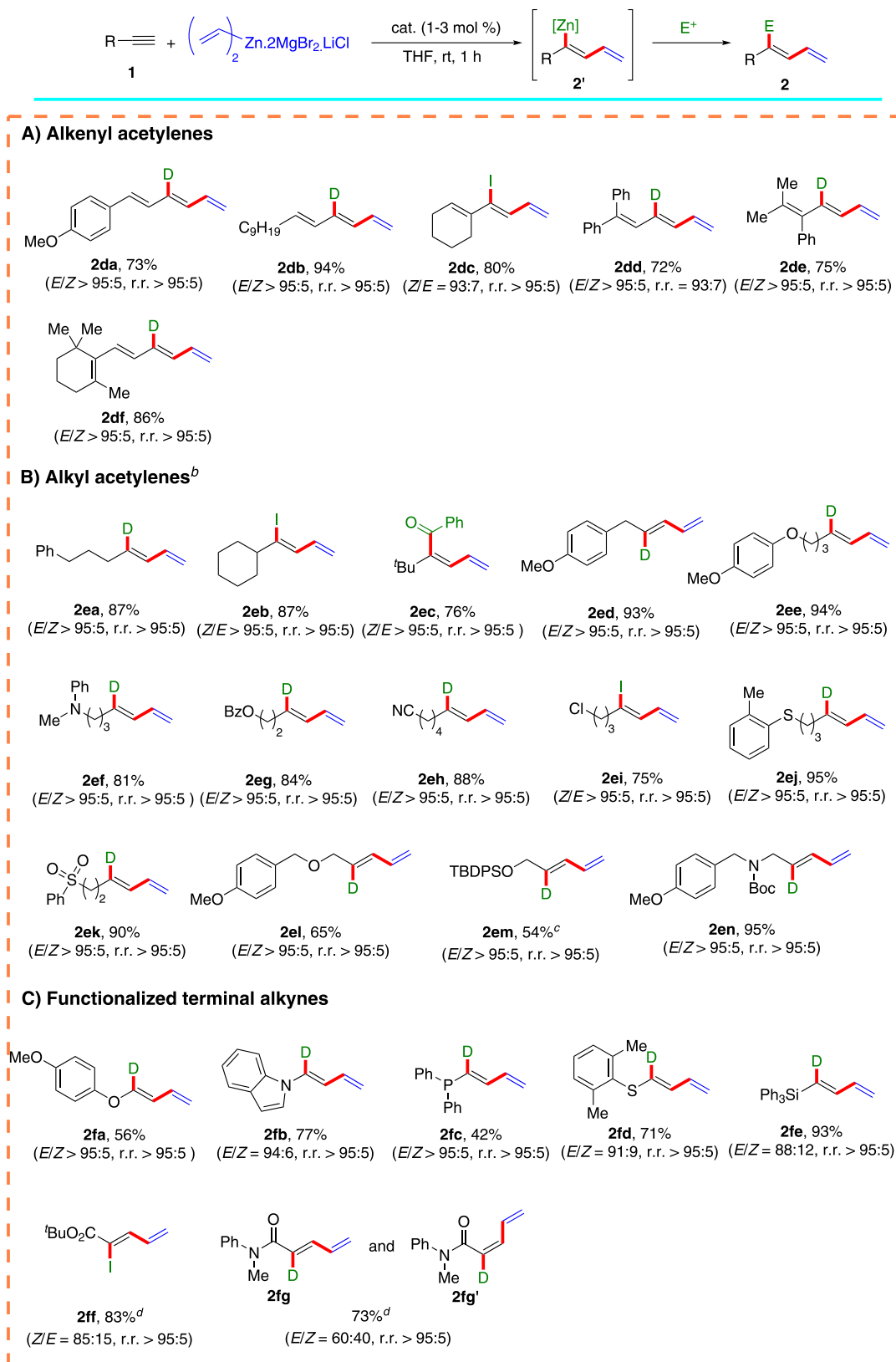
products, along with alkyne polymerization byproducts and unidentified byproducts (entry 3). We evaluated a number of other catalysts commonly used for the carbometallation of alkynes, but none of them showed satisfactory results (see Table S4 for details). Next, we systematically evaluated iron complexes bearing various types of ligands, including newly developed tridentate phenanthroline-imine iron complexes **C1–C6** (Table 1). These ligands were synthesized in three or four simple steps from commercially available 1,10-phenanthroline or 2,9-dichloro-1,10-phenanthroline (see SI for details). The structure of iron complex **C3** was confirmed by single-crystal X-ray diffraction. These phenanthroline-imine iron complexes showed high catalytic activity in the model reaction (>95% conversion of the substrate, entries 4–11). In particular, **C1–C3** and **C6**, which have ligands bearing a

Scheme 2. Iron-Catalyzed Alkenylzincation of Aryl Terminal Alkynes^a

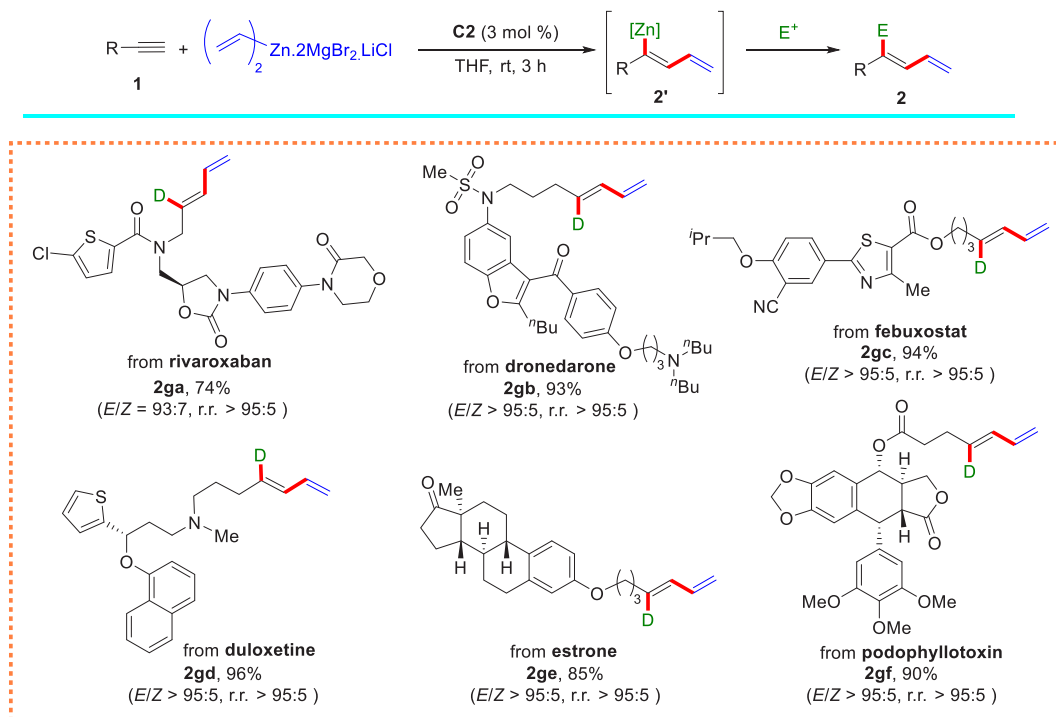
^aUnless otherwise noted, the reaction conditions were the same with those described in Table 1, entry 4 (alkyne/divinylzinc/C1 = 0.5:0.55:0.005 (mmol), in 5 mL of THF, at rt). All of the reactions exhibited full conversion of alkynes. Z/E and r.r. (regioisomer ratio) value of all the products were determined by ¹H NMR. Divinylzinc, di(prop-1-en-1-yl)zinc, and di(prop-1-en-2-yl)zinc are synthesized according to procedure A, and other zinc reagents are synthesized according to procedure E. The reactions quenched with D₂O: D₂O (150 μL), rt, 20 min see SI for details. The reactions quenched with I₂: I₂ (3.0 equiv), rt, 3 h. ^bReaction time: 10 min. ^cUsed 2.0 equiv of divinylzinc. ^dUsed 3 mol % C1 as catalyst. ^eUsed 3 mol % C4 as catalyst. ^fUsed 5 mol % C4 as catalyst.

sterically bulky 2,4,6-trisubstituted phenyl group at the 9-position of the phenanthroline moiety, gave desired vinylzincation product **2** regio- and stereoselectively in nearly quantitative yield (entries 4–8, 11). In contrast, catalyst **C4**,

which has no substituent at the 9-position, showed poor stereoselectivity (entry 9). 3,5-Di-*tert*-butylphenyl-substituted complex **C5** gave a moderate yield of **2**, along with an alkyne oligomerization or polymerization byproduct (entry 10).

Scheme 3. Iron-Catalyzed Vinylzincation of Other Terminal Alkynes^a

^aUnless otherwise noted, the reaction conditions were the same with those described in Table 1, entry 4 (alkyne/divinylzinc/C1 = 0.5:0.55:0.005 (mmol), in 5 mL of THF, at rt). All of the reactions exhibited full conversion of alkynes. *Z/E* and *r.r.* (regioisomer ratio) value of all the products were determined by ¹H NMR. The reactions quenched with D₂O: D₂O (150 μL), rt, 20 min. The reactions quenched with I₂: I₂ (3.0 equiv), rt, 3 h. The reactions quenched with benzoyl chloride: benzoyl chloride (2.5 equiv), CuCN (30 mol %), rt, 10 h. ^bUsed 3 mol % C2 as catalyst and 3 h of reaction time to alkyl acetylenes. ^cUsed 3 mol % C3 as catalyst. ^dUsed 1 mol % C2 as catalyst.

Scheme 4. Late-Stage Modification of Pharmaceuticals^a

^aReaction conditions: alkyne/divinylzinc/C2 = 0.5:0.55:0.01 (mmol), in 5 mL of THF, at rt. All of the reactions exhibited full conversion of alkynes. Z/E and r.r. (regioisomer ratio) value of all the products were determined by ¹H NMR.

Although iron complexes bearing pyridine diimine ligands (C7 and C8) showed good activity, reactions catalyzed by these complexes were not stereoselective (entries 12 and 13). Bidentate nitrogen ligands (entries 14–17), other tridentate ligands (see Table S3 for details), and monophosphine and diphosphine ligands (Table S3) were less active than the phenanthroline-imine ligands. The yield and selectivity of the reaction dropped considerably when the central metal was changed from iron to a different metal (see Table S4 for details). We also determined the purity of the catalysts C1–C3 and eliminated the influence of catalyst impurities by means of a series of control experiments (see Tables S2 and S4 for details).

Using the optimal conditions (Table 1, entry 4), we investigated the substrate scope (Schemes 2–4; note that the reactions were generally quenched with D₂O, but I₂ or benzoyl chloride was used for products with low boiling points). First, we studied the effects of substituents on the benzene ring of aryl acetylenes 1 (Scheme 2). The reaction showed good functional group compatibility: methoxy (2ac), amino (2ad, 2aw), silyl (2ae), methylthio (2af), ester (2ag), acyl (2ah), acylamino (2ai), cyano (2aj), sulfonyl (2ak), acetal (2al), trifluoromethyl (2am), and halogen (2an–2ar, 2av) substituents were tolerated. Alkyne substrates with reactive protons (–NHMe, –OH) smoothly gave desired products 2aw and 2ax in 96% and 66% isolated yields, respectively. Aryl acetylenes with an electron-donating or electron-withdrawing group on the benzene ring were all highly reactive, affording vinylzincation products 2aa–2aq in high yields under the standard conditions; however, reactions of substrates containing electron-withdrawing groups (e.g., ester, carbonyl, cyano, trifluoromethyl, and fluoro) were exhibited slightly lower stereoselectivities (2ag, 2ah, 2aj, 2am, 2an). The position of

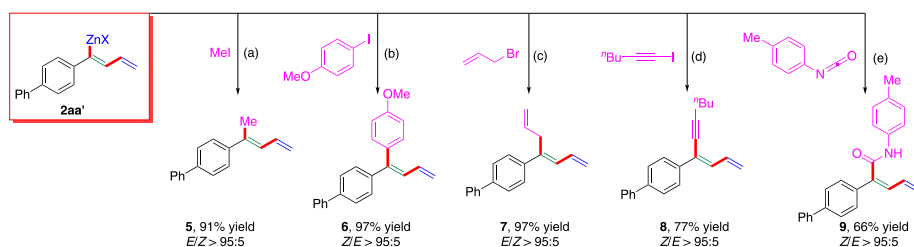
the substituent had little effect on the reaction outcome; substrates with an ortho- or meta-substituent gave excellent yields of the corresponding products (2ar–2av). In addition to phenylacetylene and its derivatives, a fused-ring-substituted acetylene (i.e., naphthyl acetylene) gave 2ba, and furanyl, thienyl, pyridyl, and indolyl acetylenes gave the corresponding product (2bb–2be). Subsequently, we evaluated reactions with a few additional alkenylzinc reagents. When C1 was used as the catalyst, β-alkyl-substituted vinylzinc reagents gave single *cis*-addition products (2ca and 2cb) in high yields. Moreover, the reaction of an α-alkyl-substituted vinylzinc reagent gave the corresponding product (2cc) in moderate yield when C4 was used as the catalyst. Unfortunately, reaction of a relatively bulky dicyclohexenylzinc reagent was sluggish, affording only a 17% yield of the desired alkenylzincation product (2cd), with oligomerized or polymerized alkyne as the major byproduct, even when 5 mol % C4 was used as the catalyst. Under the standard conditions, aryl-substituted zinc reagents gave less than 10% yields of alkenylzincation products with poor selectivity and substrate conversion (data not shown).

The reaction could be extended to alkenyl acetylenes (Scheme 3A). Both an aryl- and an alkyl-substituted enyne afforded the corresponding conjugated trienes (2da and 2db) with high yields and selectivities. Satisfactory results were also obtained with polysubstituted conjugated enynes (2dc–2df). In particular, conjugated polyene 2df, which has potential utility for synthesis of vitamin A and derivatives (vide infra), could be obtained from the corresponding conjugated dienyl acetylene in high yield with >95:5 E/Z selectivity.

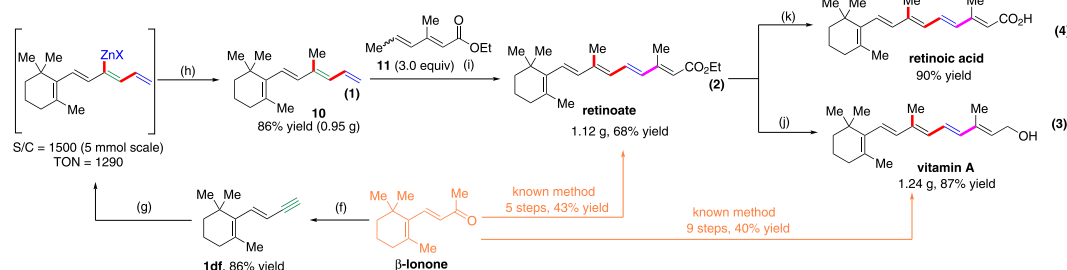
Alkyl acetylenes also smoothly underwent vinylzincation reactions in the presence of iron catalyst C2 to afford the desired products with high yields and regioselectivities (Scheme 3B). Vinylzincation products 2ea–2ed could be

Scheme 5. Applications of the Iron-Catalyzed Vinylzincation Protocol^{4a}

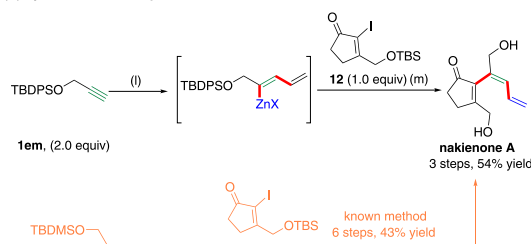
(A) C–Zn bond transformations of the products



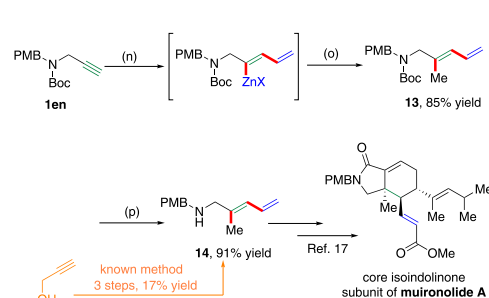
(B) Synthesis of retinoids derivatives and vitamin A



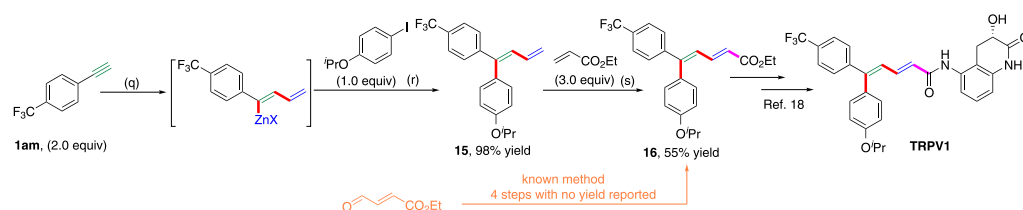
(C) Synthesis of natural product nakienone A



(D) Prepare a key intermediate in the synthesis of muironolide A



(E) Prepare a key intermediate in the synthesis of TRPV1



^{4a}Reaction conditions: (a) $\text{Pd}(\text{PPh}_3)_4$, THF, rt, 12 h; (b) $\text{Pd}(\text{PPh}_3)_4$, THF, rt, 12 h; (c) CuCN , THF, rt, 4 h; (d) CuCN , THF, -65°C , 18 h; (e) THF, 50°C , 8 h; (f) $\text{LDA}/(\text{EtO})_2\text{P}(\text{O})\text{Cl}$, THF, -78°C , 4 h; (g) divinylzinc/**C1**, THF, rt, 4 h; (h) $\text{Pd}(\text{PPh}_3)_4/\text{MeI}$, THF, rt, 12 h; (i) Grubbs-Hoveyda second generation catalyst, toluene, rt, 24 h; (j) DIBAL-H , THF, -78°C , 2 h; (k) KOH , $\text{EtOH}/\text{H}_2\text{O}$, 50°C , 4 h; (l) divinylzinc/**C3**, THF, rt, 1 h; (m) (1) $\text{PdCl}_2[(2\text{-furyl})_3\text{P}]_2/\text{tBuLi}$, DMF, rt, 24 h, (2) TBAF , THF, 0.5 h; (n) divinylzinc/**C2**, THF, rt, 1 h; (o) $\text{Pd}(\text{PPh}_3)_4/\text{MeI}$, THF, rt, 12 h; (p) $\text{CF}_3\text{CO}_2\text{H}$, DCM, rt, 24 h; (q) divinylzinc/**C1**, THF, rt, 1 h; (r) $\text{Pd}(\text{PPh}_3)_4$, THF, rt, 12 h; (s) Grubbs-Hoveyda second generation catalyst, DCM, rt, 16 h.

obtained from the corresponding primary, secondary, and tertiary alkyl acetylenes and benzylacetylene, respectively, in excellent yields and selectivities, indicating that this catalytic system was insensitive to the steric bulk of the alkyl moiety. The presence of functional groups on the alkyl moiety (i.e., ether, amine, ester, cyano, halogen, sulfide, and sulfonyl groups) had little effect on the yield or regioselectivity (**2ee–2ek**). Even highly active propargyl ethers and a propargyl amine afforded moderate to high yields of the desired products (**2el–2en**) with excellent selectivities.

Unexpectedly, even O-, N-, P-, S-, and Si-substituted acetylenes, which are typical functionalized alkynes used in organic synthesis, were also suitable substrates, affording products **2fa–2fe** in moderate to good yields (Scheme 3C). It is particularly noteworthy that electron-deficient alkynes

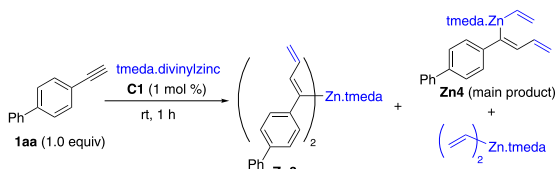
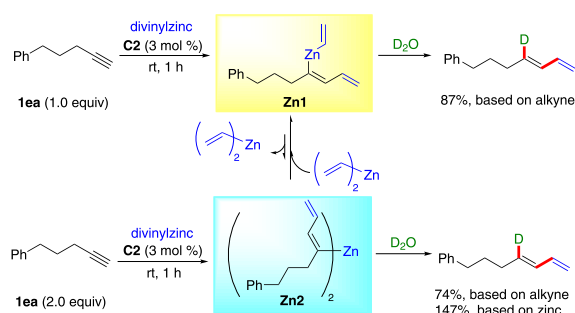
bearing an electron-withdrawing ester or acylamino group directly attached to the C–C triple bond afforded carbocation products **2ff–2fg** under the standard conditions, although the stereoselectivities were only moderate.

Because the reaction showed good functional group tolerance, we evaluated its utility for late-stage modification of alkyne-bearing multifunctionalized pharmaceutical molecules (Scheme 4). We found that rivaroxaban, dronedarone, febuxostat, duloxetine, estrone, and podophyllotoxin derivatives gave corresponding vinylzincation products **2ga–2gf** in good to high yields with excellent selectivities.

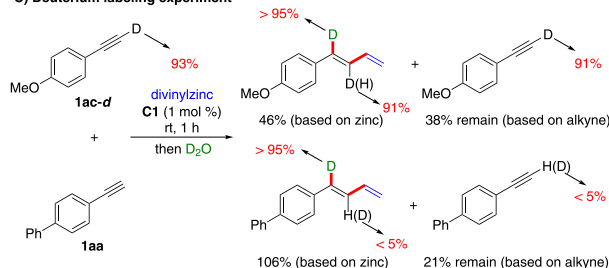
Because of the reactivity of the C–Zn bond, we expected the conjugated alkenylzinc compounds produced by this vinylzincation method to be useful intermediates for the synthesis of substituted conjugated dienes and polyenes, which are

Scheme 6. Mechanistic Study

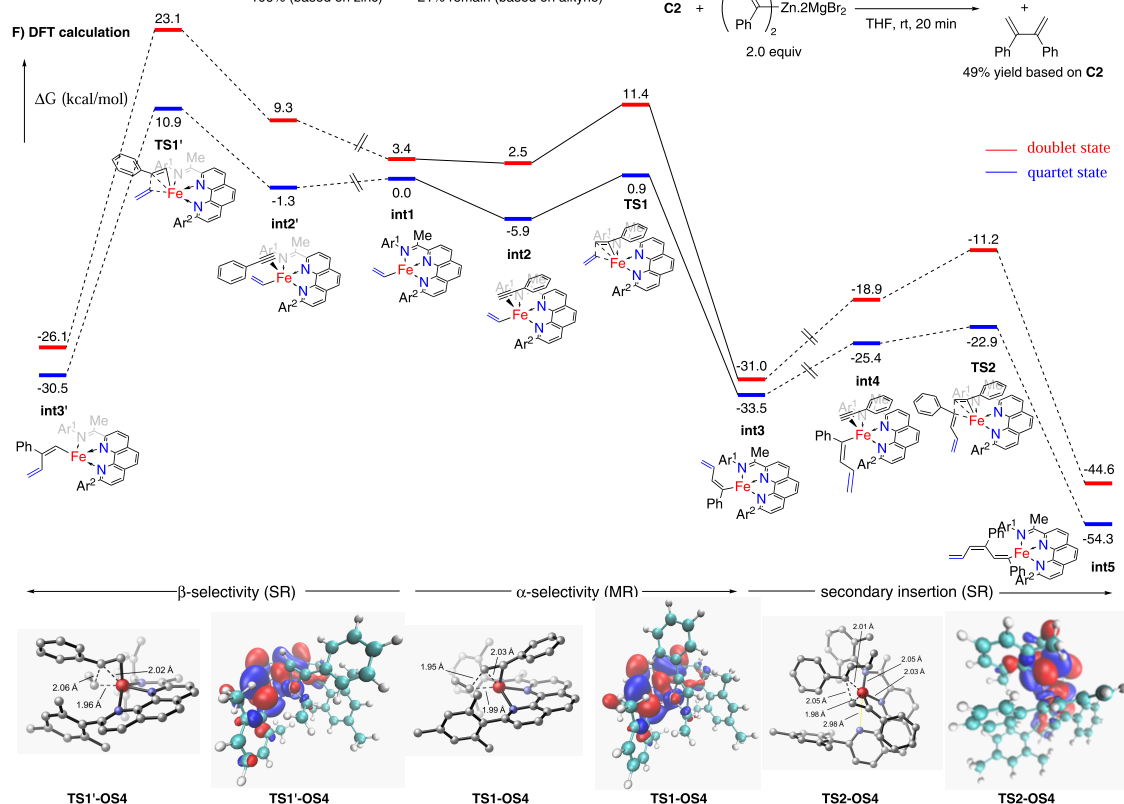
A) The structure analysis of conjugated dienylyzinc products



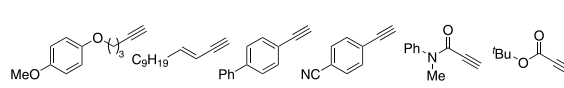
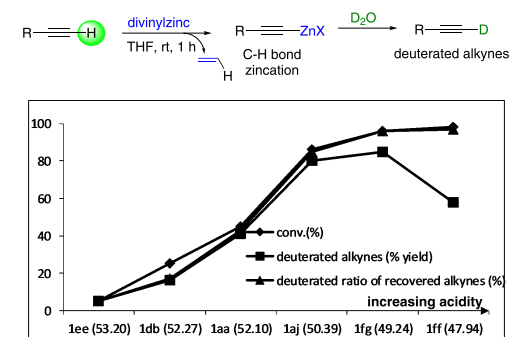
C) Deuterium labeling experiment



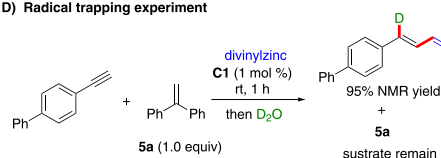
F) DFT calculation



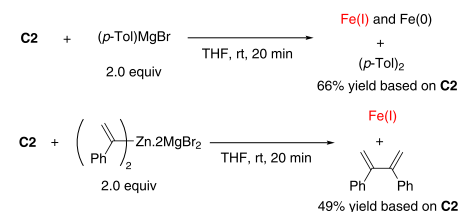
B) Studies on background reaction



D) Radical trapping experiment



E) Stoichiometric reduction experiment



widely used in the synthesis of pharmaceuticals and natural products. First, diverse transformations of the C–Zn bond of **2aa'** in situ prepared through the iron-catalyzed vinylzincation were performed (Scheme 5A). Negishi coupling reactions of

2aa' with iodomethane and *p*-methoxyiodobenzene afforded **5** and **6** in 91% yield and 97% yield, respectively, with retained configuration. Transformations of zinc group of **2aa'** to allyl and alkynyl groups mediated by copper salt ran smoothly to

afford functionalized conjugated olefins **7** and **8** in 97% yield and 77% yield, respectively. The addition reaction of **2aa'** with isocyanate produced **9** in 66% yield. Next, we carried out a series of reactions to demonstrate the applications of our method (Scheme 5B–E), starting with a synthesis of vitamin A; this compound and its derivatives display a variety of biological activities and are widely used in the field of medicine. Vinylzincation of **1df** followed by a one-pot Negishi coupling reaction with methyl iodide gave key conjugated polyene intermediate **10** as a single stereoisomer. When the vinylzincation reaction was carried out at a substrate/catalyst ratio of 1500, **10** was obtained in 86% yield (turnover number, TON = 1290; Scheme 5B-1), which demonstrates the potential utility of our protocol for the preparation of pharmaceuticals. A gram-scale cross-metathesis reaction between **10** and (2E,4E/Z)-3-methylhexa-2,4-dienoate (**11**) afforded retinoate in 68% yield with excellent stereoselectivity (Scheme 5B-2), and retinoate (1.12 g) could be easily reduced to vitamin A (Scheme 5B-3) or hydrolyzed to afford retinoic acid (Scheme 5B-4) in 87% and 90% yields, respectively. In contrast, the known method for vitamin A and its derivatives required 5 linear steps in 43% yield¹⁴ or 9 linear steps in 40% yield¹⁵ from β -ionone using equivalent hardly available zirconium reagent, respectively.

Subsequently, we used the vinylzincation method to synthesize the natural product nakienone A, a conjugated polyene that displays cytotoxicity and antibacterial activity (Scheme 5C). Specifically, catalytic vinylzincation of alkyne **1em** followed by a Negishi coupling reaction with substituted cyclopentenone iodine **12** and removal of the silyl protecting group readily gave nakienone A under mild conditions in 54% overall yield. By comparison, the known method cost much more steps for synthesis of this natural product (6 steps, 43% overall yield).¹⁶

Moreover, we used the protocol for the synthesis of compound **14**, a precursor to the isoindolinone subunit of the natural product muironolide A.¹⁷ Compound **13** was obtained in 85% yield by means of a one-step catalytic vinylzincation of **1en** followed by coupling of the product with methyl iodide in situ, and **13** could be easily transformed to **14** (Scheme 5D) by removal of the *t*-butoxycarbonyl protecting group in 91% yield. The known method to synthesize **14** from propargyl alcohol obtained only 17% overall yield.¹⁷ Finally, alkyne **1am** could be converted to diene **15** in 98% yield via a two-step procedure followed by a cross-metathesis reaction with ethyl acrylate to afford **16** in 55% yield, which is a key intermediate in the synthesis of a TRPV1 antagonist¹⁸ (Scheme 5E). This protocol is also superior to the known method from a step economy perspective.¹⁸

We performed a set of control experiments to understand the mechanism of these iron-catalyzed alkyne vinylzincation reactions. First, we characterized the structure of vinylzincation product (Scheme 6A). The ¹H NMR spectra of the product of the reaction of 1.0 equiv of alkyne **1ea** with divinylzinc showed not the expected product **Zn1**, but dienylzinc **Zn2**. We deduced that there is a dynamic equilibrium between **Zn1** and **Zn2** and that because divinylzinc has a lower boiling point than **Zn1**, it is removed during the workup, which shifts the equilibrium toward **Zn2**, which is thermodynamically stable. When excess alkyne (2.0 equiv) was used, the reaction afforded only **Zn2**, which indicates that **Zn1** could react with the alkyne to generate **Zn2**. We next performed the reaction of TMEDA-divinylzinc with higher boiling point and alkyne **1aa** and

observed that the **Zn4** was formed as the main product through ¹H NMR analysis (Scheme 6A). This experiment further supports the above deduction.

Next, we carried out reactions of terminal alkynes and divinylzinc in the absence of catalyst (Scheme 6B). These reactions revealed that as the acidity of the terminal H of the tested alkynes increased, the rate of zincation of the terminal alkyne C–H bond increased substantially, and other side reactions were also coupled to highly active functionalized alkynes. These experiments indicate that the iron catalysts promoted vinylzincation efficiently enough to suppress background C–H zincation.

We also conducted a deuterium-labeling experiment involving a 1:1 (mol/mol) mixture of alkynes **1ac–d** and **1aa** and divinylzinc under the standard reaction conditions (Scheme 6C). The products and any remaining substrates were separated and identified after the reaction was quenched with D₂O. The fact that no H–D exchange occurred during the reaction indicates that the iron catalysts did not react with the terminal C–H bond of the alkynes. The addition of the radical scavenger 1,1-diphenylethylene had no effect on the reaction, and the scavenger was completely recovered, which suggests that the reaction did not proceed via a radical process (Scheme 6D).

Using known methods,^{13b,19} we carried out an equivalent reduction experiment to determine the oxidation state of the active catalyst (Scheme 6E). When (*p*-Tol)MgBr was used as a reducing agent, 4,4'-dimethyl-1,1'-biphenyl was obtained in 66% yield (based on **C2**), which indicates Fe(I) was the main iron species that formed, along with a small amount of Fe(0), while using di(1-phenylvinyl)zinc as a reducing agent, 2,3-diphenylbutadiene was obtained in 49% yield (based on **C2**), which indicates only Fe(I) (apparent valence state) species was formed. The active iron catalyst prepared by this method showed the same activity as that in situ generated one in the catalytic vinylzincation of **1aa** (see Table 1, compare entry 6,7 to entry 5). We measured the in situ prepared active iron catalyst with EPR at 293 K and observed a clear radical signal (Figure S1), which indicates that an open shell iron catalyst was formed.

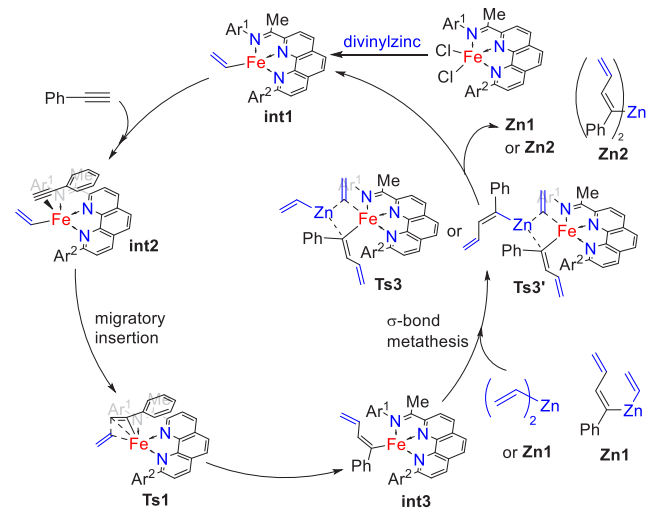
We used density functional theory calculations to explain the regioselectivity and chemoselectivity in both high-spin ($S = 3/2$) and low-spin ($S = 1/2$) states at the (dispersion-corrected) unrestricted ω B97XD/6-31G*/TZVP level of theory with CPCM-U ω B97XD/def2-TZVPP single-point energy calculations (see Table S7 for details). The calculations showed that the activation energy of Fe(I) in a low-spin state (doublet state, activation energy is 8.0 kcal/mol) was 7.1 kcal/mol higher than that of high-spin Fe(I) (quartet state, activation energy is 0.9 kcal/mol) and also showed that the active iron species was always in a quadruple state and that no spin crossover occurred during the catalytic process (Scheme 6F). The energy of α -addition intermediate **int2** was about 4.6 kcal/mol lower than that of β -addition intermediate **int2'** and the transition state of vinyl migration insertion **TS1'** (10.9 kcal/mol) had a higher energy barrier than **TS1** (0.9 kcal/mol). In general, the energy difference between transition states **TS1** and **TS1'** (activation energy), which lead to products arising from α - and β -selectivity, respectively, was very large (10.0 kcal/mol), indicating good regioselectivity; this result is consistent with the experimentally observed result that only α -addition products were obtained. Because of the large steric bulk of the catalyst, the energy of **int4** was 8.1 kcal/

mol higher than **int3**, and the activation energy (10.6 kcal/mol) for secondary insertion was much higher than that for primary insertion (0.9 kcal/mol). Examination of the catalyst structure showed that the length of the Fe–N₃ bond in secondary insertion transition state **TS2** (2.98 Å) was much larger than the bonding range (Fe–N coordination bond lengths are generally 1.8–2.3 Å).²⁰ In addition, the large steric bulk of the ligand and the conjugated dienyl resulted in **int4** having a crowded reaction pocket, which also indicates that insertion of the alkyne into the Fe–C bond via transition state **TS2** was difficult. Generally speaking, the rate of the secondary addition reaction was much lower than that of the primary addition reaction, thus accounting for the high chemoselectivity.

By analyzing the transition state structures, we found that the distance between the iron atom and the phenanthroline plane in each transition state was different; specifically, the distance in **TS1** was much smaller than the distances in the other two transition states ($= 0.60$ Å, $d_{TS1'} = 0.78$ Å, $d_{TS2} = 0.91$ Å), which means that the iron atom d-orbital and the phenanthroline π -orbital had different degrees of overlap. To confirm this, we simulated the frontier molecular orbitals of the three transition states and found that the Fe d_{z^2} -orbital in **TS1** and the phenanthroline π -orbital had strong p– π conjugation effects, which dispersed the charge in **TS1** and lowered its energy relative to those of the other transition states. These results account for the high yield, regioselectivity, and chemoselectivity of the reaction.

On the basis of the above experimental results and DFT calculations, we could draw the whole catalytic cycle as shown in Scheme 7. First, precatalyst **C1** is reduced to the active

Scheme 7. Proposed Catalytic Cycle



species **int1** by divinylzinc. The **int1** coordinates an alkyne to form **int2**, which undergoes migration insertion through **TS1** to generate **int3**. Then **int3** undergoes transmetalation with divinylzinc via σ -bond metathesis (**TS3**) to afford the product **Zn1**. When the alkyne is excessive (the divinylzinc is a limiting reagent), the **int3** may undergo transmetalation with previously formed **Zn1** via **TS3'** to generate dienylynzinc **Zn2**. Even still, the other catalytic cycles based on Fe(0) or Fe(II) cannot be definitively excluded at the current stage.

CONCLUSION

In conclusion, we have realized the first method for highly selective vinylzincation reactions of terminal alkynes by using new iron catalysts bearing tridentate 1,10-phenanthroline-imine ligands. The substrate scope was broad, and the method showed good functional group compatibility and excellent chemoselectivity, regioselectivity, and stereoselectivity. A variety of natural products and pharmaceutical intermediates could be efficiently synthesized by simple transformations of the conjugated alkenylzinc products. Mechanistic experiments and density functional calculations showed that the iron-1,10-phenanthroline-imine catalysts efficiently promoted transfer of the vinyl group to the alkynes. No substitution reactions between the zinc reagents and the terminal C–H bond of the alkynes was observed, and the crowded cavity of the catalysts hindered further reaction of the product. This work not only markedly expands the types of organozinc reagents, but also can be expected to inspire the synthesis of other organo-metallic reagents and the development of other iron-catalyzed reactions.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.1c11072>.

Supporting methods and data (PDF)

Accession Codes

CCDC 2100653 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

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